

PROJECT TITLE

HUMAN EMOTION DETECTION FROM VOICE

Abstract

This project develops a real-time Speech Emotion Recognition system to classify human emotions from audio input. Using the RAVDESS dataset, we extracted acoustic features like MFCC, Chroma, and Mel-spectrograms with Librosa. A Random Forest classifier trained via Scikit-learn achieved ~ 70% accuracy across 8 emotions: neutral, calm, happy, sad, angry, fearful, disgust, and surprised. A Streamlit web interface allows users to upload audio files and receive instant emotion predictions with confidence scores and probability graphs. The system demonstrates the practical application of machine learning in affective computing for human-computer interaction.

1. Introduction

Human communication relies heavily on emotional context conveyed through speech. Speech Emotion Recognition (SER) aims to identify emotional states from vocal cues, enabling more empathetic AI systems. This project addresses the challenge of automated emotion detection using machine learning. The objective is to build an end-to-end system that takes raw audio, processes it to extract relevant features, and classifies the speaker's emotion in real time. Applications include mental health monitoring, customer service analysis, and interactive voice assistants. We use the RAVDESS dataset for its high-quality, labeled emotional speech samples to ensure robust model training.

2. Tools Used

Tool/Library	Purpose
Python 3.9	Core programming language
Librosa	Audio loading and feature extraction: MFCC, Chroma, Mel
Scikit-learn	Model training: RandomForestClassifier, StandardScaler
Streamlit	Web UI for audio upload and result visualization
Pandas/Numpy	Data handling and numerical operations
Joblib	Model serialization for saving/loading
RAVDESS Dataset	1440 audio files with 8 emotional labels

3. Steps Involved in Building the Project

Step 1: Dataset Preparation

Downloaded the RAVDESS dataset containing 24 actors speaking 2 statements with 8 emotions. Files were organized by actor and emotion label extracted from filenames.

Step 2: Feature Extraction

For each audio file, 180 features were extracted using Librosa: 40 MFCCs to capture timbral texture, 12 Chroma features for pitch class profiles, and 128 Mel-spectrogram coefficients for frequency content. Features represent the statistical mean of each coefficient over time.

Step 3: Model Training

Data was split 80/20 for training/testing. Features were normalized using StandardScaler. A RandomForestClassifier with 200 estimators was trained on the processed data. The model and scaler were saved as .pkl files using Joblib.

Step 4: UI Development

Built a Streamlit application with file uploader supporting WAV/MP3 formats. The app loads the trained model, processes uploaded audio through the same feature pipeline, and runs prediction.

Step 5: Prediction & Visualization

The system outputs the predicted emotion, confidence percentage, and a bar chart of probabilities for all 8 emotions. Temp files are used for processing and auto-deleted post-prediction.

Step 6: Testing & Validation

Tested with unseen RAVDESS files and live recordings. Achieved ~ 70% test accuracy. Angry and happy emotions showed highest precision due to distinct acoustic signatures.

4. Conclusion

The Speech Emotion Detector successfully classifies 8 emotions from voice with satisfactory accuracy using classical ML techniques. The project demonstrates that MFCC and spectral features combined with Random Forest provide a lightweight, effective solution for SER without deep learning. The Streamlit UI makes the model accessible for real-time demos. Limitations include reduced accuracy on noisy audio and speaker dependency. Future work includes implementing LSTM/CNN models for improved temporal analysis and adding live microphone recording. This system forms a foundation for integrating emotional intelligence into conversational AI and assistive technologies.